

Geo-spatial Learning: Can *additional modalities* improve cross-view localisation?

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Can we improve cross-view geo-localisation (ground→satellite retrieval) by narrowing the satellite search space via a GPS prediction from a ground image? — coupling Sample4Geo with GeoCLIP

— DATASET

MMLandmarks

Open-source multimodal geo-spatial benchmark of US landmarks with one-to-one correspondence across four modalities:

Ground Images
Satellite
Wikipedia Text
GPS Coords

17k
Train landmarks

311k
Train ground imgs

187k
Satellite imgs

1k
Query set

18k
Query Images

Both
Text and GPS for all landmarks

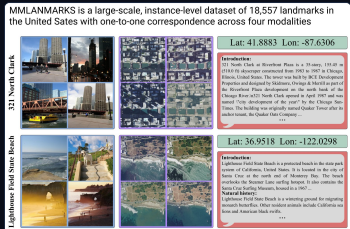


Fig 1. MMLandmarks — each landmark provides ground photos, satellite imagery, GPS and Wikipedia text across all 18,557 US sites.

— PROBLEM

Cross View Retrieval

Given a **ground-level photo**, retrieve the matching **satellite image** from a gallery of ~100k candidates. The two views look completely different — the model must learn a shared embedding where same-location pairs are close.

Key question: Can narrowing the satellite space using GPS as an intermediate modality improve retrieval, since the model can focus on more finegrained detail

— MOTIVATION

Why GPS as a Bridge?

For practical scenarios, *satellite images usually have GPS metadata*, but a GPS coordinate doesn't guarantee a paired satellite image. This motivates our two-stage approach

Since our new approach also restricts what Sample4Geo can learn, having GeoClip as a sort of upper bound, we wanted to improve upon Geoclip too. Each landmark has on average 18 images, but it varies. This motivated us to try and create a version of Geoclip able to take multiple images for one landmark.

— BASE ARCHITECTURES

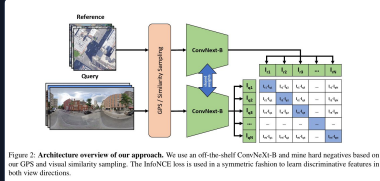


Fig 2. **Sample4Geo** — shared ConvNeXt-B encoder for ground & satellite. Hard negatives via GPS proximity then dynamic similarity sampling. InfoNCE loss in both directions.

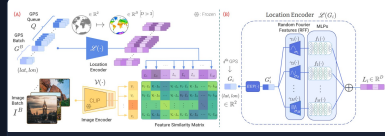


Fig 3. **GeoCLIP** — frozen CLIP image encoder with a finetuned MLP on the end, paired with a location encoder (Random Fourier Features + MLPs). Contrastive loss aligns image embeddings to GPS coordinates.

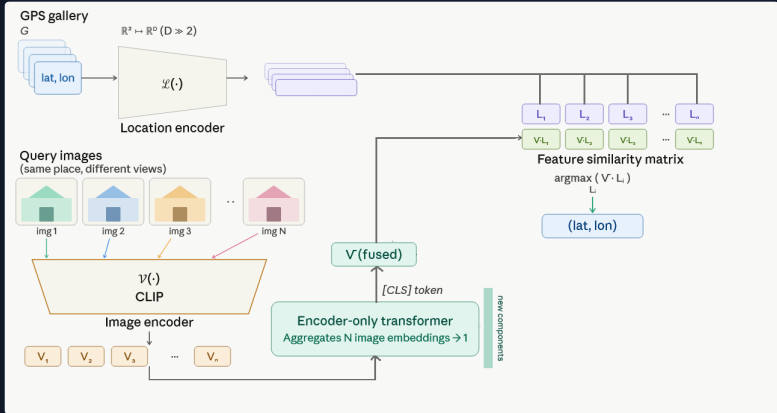
— HYBRID PIPELINE - GPS-GUIDED CROSS-VIEW RETRIEVAL



— EXPERIMENTS & RESULTS

MODEL					
— GEOCLIP —	ACC@1KM	ACC@25KM	ACC@200KM	MEDIAN-ERR	
Zero-shot	6.67%	28.79%	44.48%	294km	No MMLandmarks ft
Multi-image weighted mean	12.13%	40.67%	58.45%	117km	Softmax CosineSim mean
Multi-image Attention	13.32%	54.46%	72.63%	109km	Encoder-only transformer
— SAMPLE4GEO —	R@1	R@5	R@10	MAP@1K	
ConvNeXt zero-shot	0.25%	0.76%	1.13%	0.59%	ImageNet only
v2 (224px, ep30)	7.21%	18.52%	25.31%	13.10%	InfoNCE + DSS
v3 (384px, ep36)	8.58%	18.13%	22.29%	13.25%	Multi-pos. loss
— HYBRID PIPELINE —	R@1	R@5	R@10	MAP@1K	
ZS GeoCLIP + v3	10.20%	21.11%	26.80%	15.84%	Finetuned Sample4Geo
Finetuned Hybrid	13.80%	22.23%	25.41%	17.52%	
★ Hybrid w/ Attn GeoCLIP	15.50%	19.86%	25.20%	45.00%	Best model

— NEW GEOCLIP ARCHITECTURE



— KEY FINDINGS

What We Learned

- Domain training for Sample4Geo.**
Zero-shot: 0.25% R@1. Fine-tuned: 8.58%
- Resolution is important v3 (384px) beats v2 (224px) by +1.37% R@1.** Fine-grained context improves performance
- CLIP backbone dominates GeoCLIP**
Without additional changes to GeoClip architecture, finetuning resulted in negligible improvement.
- Masked Multi-head Attention did not** improve performance significantly over a weighted mean, which was unexpected.
- GPS narrowing worked** Hybrid pipeline saw a ~50% improvement on Sample4Geo R@1. Multi-image GeoClip improved even further

— METRIC DISCUSSION

Per-Landmark Evaluation

Unaveraged evaluation treats each of the ground images for a single landmark individually, weighing landmarks with more images heavier in the final statistics. This might cause better known landmarks to bias the statistics compared to real performance.

For this poster we show all statistics in this way, since it was the way the MMLandmarks paper did it (we ran out of time). For our final poster, we hope to report everything in per landmark statistics

— DISCUSSION

What We'd Try Next

- Out-of-domain evaluation on Uni-1652 to test generalisation.
- Encode Wikipedia text as a third contrastive modality — text may help the model learn during training
- Adaptive GPS radius based on GeoCLIP confidence instead of fixed 20 km.